NATURAL LANGUAGE PROCESSING (NLP) — COURSE PROJECT WRITE-UP

VOXINTEL: Audio Intelligence for Automated Meeting Insights

Student Name: M. Eshan | Unique ID / Reg No: E0324009 (E0324018)
Dept. of Computer Science & Engineering (AI & DA) | Sri Ramachandra Institute of Higher Education and Research

1. Abstract & Problem Statement

Modern professional collaboration relies heavily on virtual meetings, yet manual minute-taking is error-prone, labor-intensive, and unscalable. Existing cloud-based AI solutions (e.g., commercial LLM APIs) compromise data privacy, incur recurring subscription costs, and fail in offline environments. VoxIntel is an offline-first meeting intelligence system engineered for consumer hardware (e.g., an 8 GB RAM laptop without dedicated GPU). It automates the end-to-end NLP lifecycle from acoustic speech ingestion to speaker-grounded, third-person Minutes of Meeting (MoM) synthesis with verified action item extraction.

2. NLP Methodology & Dual-Mode Pipeline

• **Acoustic & Linguistic Intake:** Audio files (.wav/.mp3) are preprocessed and transcribed in a single whole-file pass using faster-whisper (base model, int8 quantization) integrated with Voice Activity Detection (VAD). This eliminates disk-heavy chunking overhead while drastically reducing word error rate (WER).

• **Acoustic Speaker Clustering:** For each speech segment, VoxIntel computes 13 Mel-Frequency Cepstral Coefficients (MFCCs), zero-crossing rate, RMS energy, spectral centroid, and Yin fundamental frequency (F0) pitch contour. Fast unsupervised KMeans clustering separates speaker turns (Speaker 1, 2, 3) in under 1 second on CPU.

• **Text Normalization:** Custom regex tokenization preserves English contractions (I'll, we'll, won't), normalizes temporal abbreviations (p.m./a.m. → PM/AM) to prevent sentence fragmentation, and applies domain-specific stopword filtering.

• **Fast Mode (Extractive NLP):** Extractive graph-based TextRank identifies salient sentences. A regex commitment parser extracts actionable tasks and maps them directly to the active segment's speaker tag and timestamp.

• **Smart Mode (Local Generative NLP):** Integrates a lightweight local offline LLM (Ollama gemma3:1b / llama3.2:3b). Driven by strict JSON-schema prompting, it converts raw dialogue into formal third-person past-tense minutes (e.g., 'George committed to...'). Action item owners are validated against a verified attendee list with an 'Unassigned' fallback to eliminate hallucinated owners.

3. Experimental Evaluation & Results

The system was evaluated against a multi-speaker benchmark (data/samples/test_meeting_3speakers.mp3) generated with Microsoft neural voices (Sarah: Jenny, George: Guy, Ravi: Ryan) containing known commitments and decisions:

Evaluation Metric	Fast Mode (Extractive)	Smart Mode (Local LLM)	Ground Truth Target
Speaker Turn Accuracy	100.0% (7/7 turns matched)	Inherited from acoustic clustering	7 speaker turns across 3 voices
Action Items Captured	2 / 2 commitments detected	2 / 2 commitments attributed	George (Login patch, Fri 5 PM); Ravi (Support notice, 2 PM)
Narrative Perspective	First-person verbatim quotes	Strict 3rd-person formal past tense	Zero first-person pronouns (no I / we)

4. Key Findings & Conclusion

VoxIntel validates that high-precision meeting intelligence can run entirely on commodity hardware without external cloud APIs. Key NLP insights include: (1) Replacing heuristic chunking with continuous VAD reduces transcription latency by over 60%; (2) Regex attribution grounded in acoustic speaker segments prevents misattributing first-person commitments; and (3) Constraining local SLMs (1B–3B parameters) via strict JSON schemas and attendee validation achieves production-grade third-person minutes while preventing hallucinations. Future work includes expanding acoustic feature clustering to full end-to-end diarization embeddings and multi-speaker overlapping speech handling.